

ARSLAN JAFFAR

Senior Backend Engineer — AI Systems Engineer — Distributed Systems

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PROFESSIONAL SUMMARY

Senior Software Engineer with 4+ years of experience designing scalable backend platforms, microservices, and event-driven systems using Node.js, NestJS, PostgreSQL, Redis, and RabbitMQ.

Currently focused on AI Systems Engineering including Retrieval-Augmented Generation (RAG), agent orchestration, vector databases, LLM integrations, and production AI backend architectures.

Strong expertise in system design, distributed systems, production debugging, scalability engineering, and cloud-native backend development.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, SQL
Backend	Node.js, Express.js, NestJS, Fastify
AI Systems	RAG, LangChain, LangGraph, CrewAI, MCP, Prompt Engineering
LLM Integration	Gemini, Ollama, Open-source LLMs, Function Calling
Vector Databases	Qdrant, ChromaDB
Databases	MongoDB, PostgreSQL, MySQL, Redis
Messaging	RabbitMQ, BullMQ, Kafka
Architecture	Microservices, Event-Driven Systems, CQRS, REST APIs
DevOps	Docker, AWS, GitLab CI/CD, Nginx
Performance	Caching, Query Optimization, Indexing, Profiling

PROFESSIONAL EXPERIENCE

Senior Software EngineerOct 2023 - Present

Beyond Technologies, Lahore

- Designed scalable microservices using Node.js and NestJS following SOLID architecture principles.
- Built event-driven distributed systems using RabbitMQ for asynchronous service communication.
- Implemented Redis caching reducing API latency by 35%.
- Optimized complex PostgreSQL and MySQL queries improving execution performance by 50%.
- Built Dockerized CI/CD deployment pipelines using GitLab CI.
- Designed production logging, monitoring, retry, and fault-tolerant workflows.
- Leading internal AI backend initiatives including RAG pipelines, agent workflows, vector search, and LLM orchestration.

Software Engineer (Full Stack)Feb 2022 - Oct 2023

Eventbuizz, Lahore

- Built scalable REST APIs using Node.js and Express.js.
- Designed backend services for event management platform processing large-scale user activity.
- Implemented authentication, authorization, pagination, caching, and performance optimization.
- Collaborated on frontend-backend contracts and real-time platform features.

AI ENGINEERING PROJECTS

Enterprise AI Assistant Platform

- Built multi-agent workflows using LangGraph and LangChain.
- Designed Retrieval-Augmented Generation pipeline using Qdrant vector database.
- Integrated Gemini and local LLMs using Ollama.
- Built memory-aware conversational workflows and tool-calling agents.
- Implemented MCP-based external tool integrations.

Production AI Backend Architecture

- Designed intelligent request routing between LLM, Agent, and RAG execution layers.
- Implemented chunking, embeddings, retrieval, reranking, and contextual memory.
- Built stateful agent orchestration with observability and retry mechanisms.

CORE PROJECTS

Voxtasia Audio Streaming Platform

- Built distributed backend services for real-time audio streaming.
- Implemented RabbitMQ-based asynchronous media processing.
- Used Redis caching for metadata and session optimization.
- Designed scalable APIs for concurrent traffic.

AI Fleet Management ERP

- Built enterprise-grade microservice architecture.
- Developed fleet, finance, HR, and analytics services.
- Designed event-driven communication and optimized large-scale data workflows.

EDUCATION

BS Information Technology

Virtual University of Pakistan

KEY STRENGTHS

- Production Backend Engineering
- Distributed Systems Design
- Event-Driven Architecture
- Performance Optimization
- AI Backend Infrastructure
- System Debugging and Root Cause Analysis